

Characterization of soil moisture responses on a hillslope to sequential rainfall events during late autumn and spring

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[1] In this study, the results of intensive soil moisture monitoring and analysis are reported in order to highlight the seasonal differences in hydrological processes on a hillslope scale. A soil monitoring system for a steep hillslope was designed and installed to monitor the temporal features of soil moisture with respect to the terrain attributes. Comparison of soil moisture analyses for two different periods, one during late autumn and the other during late spring, yielded different soil moisture characteristics, even though both seasons were subject to similar rainfall and saturation conditions. Comparative analyses indicated that soil moisture response patterns can be characterized and classified into three different zones. A substantial portion of the rainfall contributed to recharge of the water storage in autumn, while soil moisture responses were less variable during spring. Infiltration patterns between the two seasons were significantly different, hence the different distribution of soil moisture and hydrological responses. Vertical flux and recharge were the requirements for lateral redistribution of soil moisture in the study area. The seasonal impact of the vegetation, such as transpiration, fine root density, and interception, is apparent on the soil moisture distribution on the hillslope scale as well as runoff features in the mountainous forest catchment.

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1. Introduction

[2] Soil moisture is an important variable in determining the hydrological processes across hillslope scales [Ridolfi *et al.*, 2003; Mertens *et al.*, 2004; Zehe *et al.*, 2005; Maeda *et al.*, 2006]. Soil water content also significantly affects infiltration, evapotranspiration, solute transport, interaction with groundwater and the dynamics of the ecosystem [Rodriguez-Iturbe and Porporato, 2004; Tromp-van Meerveld and McDonnell, 2006; Broksma and Bierkens, 2007]. The dynamics of unsaturated zones and the generation of lateral subsurface stormflows are heavily dependent upon the spatial and temporal distribution of soil moisture, soil pore pressure and bedrock topography [Torres *et al.*, 1998; Uchida *et al.*, 2004; Wilson *et al.*, 2004].

[3] The temporal and spatial distribution of soil moisture has been studied across various catchment scales [Famiglietti *et al.*, 1998; Western and Grayson, 1998; Gomez-Plaza *et al.*, 2001; Qiu *et al.*, 2001; Wilson *et al.*, 2004, 2005; Maeda *et al.*, 2006; Brocca *et al.*, 2007]. Different monitoring programs with various measurement devices have been used to study soil moisture. Sampling space intervals, ranging between 0.045 and 100 m [Nyberg, 1996; Famiglietti *et al.*, 1998], and sampling frequencies vary from daily to biweekly. Sampling networks can be classified by a transect along or across the contours of a hillslope, both grid based and irregular [Nyberg, 1996; Famiglietti *et al.*, 1998; Western

and Grayson, 1998; Brocca *et al.*, 2007]. The impact of topography, soil property, radiation, and land cover on soil moisture redistribution depends upon the conditions of the study area, such as slope, relative wetness, and vegetation cover [Anderson and Kneale, 1980; Dendrou and Dendrou, 1999; Canton *et al.*, 2004; Wilson *et al.*, 2004]. The spatial and temporal distribution of soil moisture during stormflow and subsequent low-flow periods can be significantly affected by subsurface conditions governed by macropores [Beven and Germann, 1982; Noguchi *et al.*, 1997].

[4] Several studies have focused on the lag times of hydrologic mechanisms for runoff generation on a hillslope scale [Anderson and Burt, 1978; Onda *et al.*, 2001; Uchida *et al.*, 2003a; Haga *et al.*, 2005]. Previous studies used hydrometric and hydrochemical monitoring in the fields to understand the flow path and source of runoff [McDonnell, 1990; Anderson *et al.*, 1997; Asano *et al.*, 2002; Buttle *et al.*, 2004]. The role of surface and subsurface topography in the generation of subsurface stormflow [Freer *et al.*, 2002] and the water table [Blazkova *et al.*, 2002] was also studied. The response features of soil moisture during and after sequential rainfall events need to be explored in the context of ecohydrological processes such as interception, evapotranspiration, root activity, and runoff generation [Kozlowski *et al.*, 1997; Rodriguez-Iturbe and Porporato, 2004; Haga *et al.*, 2005].

[5] However, not enough studies have explored the features of soil moisture response within the context of spatial characterization and classification of seasonal differences on a hillslope scale. In order to fully understand the hydrological processes related to rainfall-runoff parameters, the response patterns of soil moisture across a hillslope need

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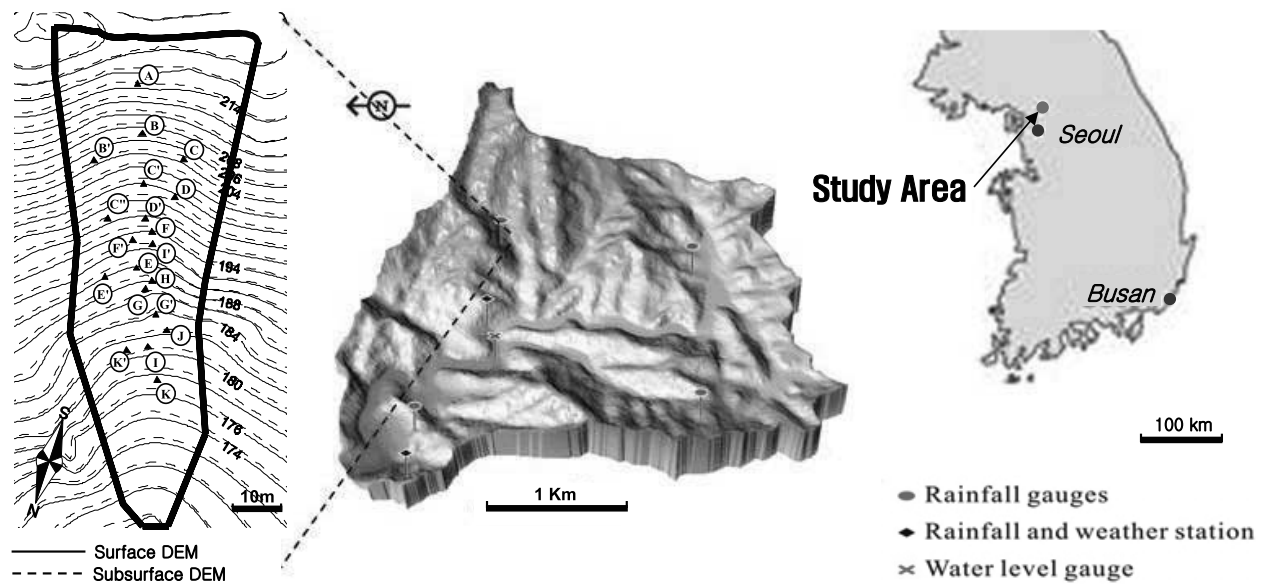


Figure 1. The location of the Sulmachun catchment and the study area with surface and bedrock topography and monitoring locations.

to be studied. In this study, a multiplex TDR system [Soilmoisture Equipment Corporation, 2003] for 20 locations was employed for intensive monitoring of soil moisture on a hillslope (3200 m²) during 6–22 November 2003 and 2–22 May 2004. These two study periods were selected because of the similar rainfall distribution and mean soil moisture conditions, which surprisingly yielded distinctly different hydrologic response patterns. Data collected as part of this study have been used to address the following questions: (1) How can the temporal responses of soil moisture during sequential rainfall events be spatially characterized and classified on the hillslope scale? (2) How are different spatial patterns of average soil moisture affected by terrain attributes between the two seasons? (3) How do the soil moisture responses differ between the two seasons in terms of variation, the response time scale, water recharge, the predictability based upon rainfall and antecedent moisture content, and the relationship between the runoff responses?

2. Methodology

2.1. Study Area

[6] Spatial and temporal variations in soil moisture were measured on a hillslope in the Sulmachun catchment, located at the headwaters of the Imjin River in northern South Korea. The longitude of the Sulmachun catchment is between 126°55'54" and 126°54'57" and the latitude between 37°54'57" and 37°56'23". Figure 1 shows the location of the Sulmachun catchment (8.5 km²) and the study area (3200 m²) on the Korean peninsula, in addition to surface and subsurface digital elevation models (DEM). The soils in the study area are Leptosols (FAO classification). The mean annual precipitation is approximately 1,500 mm. The rainy season usually starts in the middle of June and lasts until the end of

August. The mean minimum and maximum temperatures are −10° and 35°C, respectively. The study area is primarily covered by a mixture of *Polemoniales* and shrubby *Quercus*. The hillslope slope varies between 30° and 45° and consists of gneiss composites underlain by granite bedrock. Scales or crevices are minor, and connectivity among apertures is negligible [Jung, 2001]. The depth of the soil is between 30 cm and 85 cm, which was measured by manual inspection with an iron pole. Soil samples were collected for all 45 monitoring points and used for grain size analysis, from which hydraulic conductivities can be estimated [Kasenow, 2001]. The average, standard deviation, maximum, and minimum hydraulic conductivities were 3.12×10^{-3} , 2.69×10^{-3} , 1.42×10^{-2} , and 1.71×10^{-4} in m/s, respectively. The porosity was also estimated from the grain size analysis [Kasenow, 2001], and varied from 0.28 to 0.35. Massive macropores, due to the intensive development of roots, made the estimation of actual pore volume difficult. Three undisturbed soil core samples were collected in the study area. The volumetric method for estimation of porosity was employed. The porosity ranged between 45% and 48%.

2.2. Method and Hydrologic Observations

[7] Topography, soil texture distribution, vegetation, and bedrock depth are the primary components of the distribution of soil water content [Freer et al., 2002; Wilson et al., 2004]. Considering the relatively small study site, the spatial variability of the vegetation was assumed to be negligible. Particle analysis of the soil samples did not indicate any systematic difference in soil texture distribution along the hillslope. Refined DEMs, with a 0.5 m resolution for the surface and bedrock, were obtained by undertaking an intensive topographical survey of the study area using

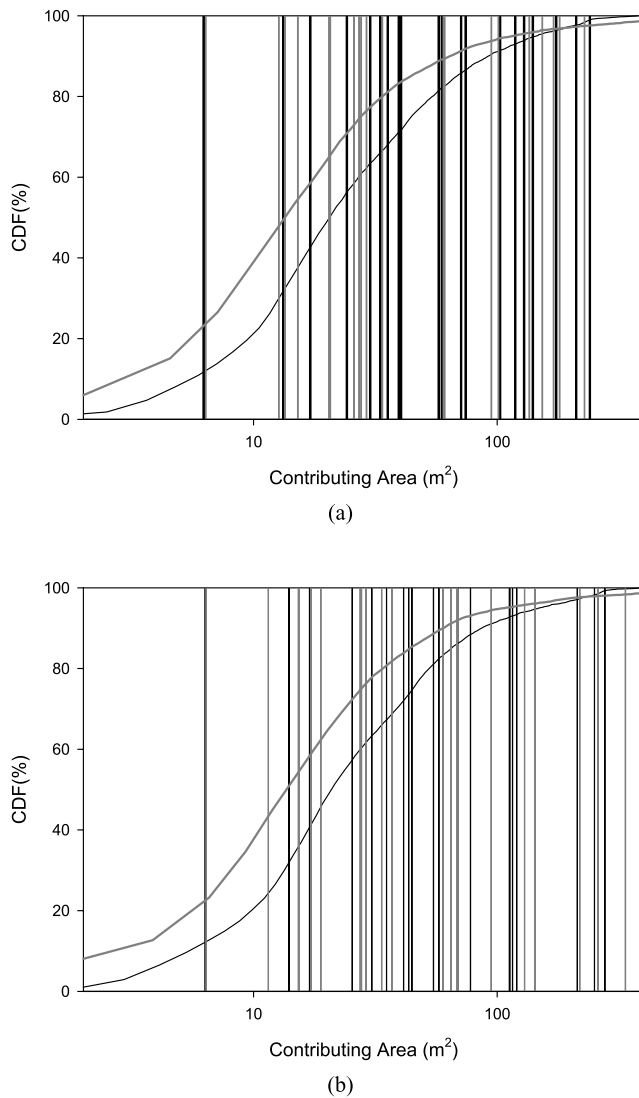


Figure 2. Cumulative frequency distributions of contributing areas using the $MD8$ algorithm (black lines) and $MD\infty$ algorithm (gray lines) using (a) surface topography and (b) bedrock topography; the vertical bars are monitoring locations with the corresponding contributing areas.

a theodolite (DT-208P, TOPCON) and iron pole (see Figure 1).

[8] The topographic wetness index [Beven and Kirkby, 1979] is used to predict the spatial patterns in a saturated area [Guntner et al., 2004] or water tables [Blazkova et al., 2002]. The upslope contributing area is computed as upstream cells are routed to downslope cells from DEM. The multiple flow direction algorithm ($MD8$) [Quinn et al., 1991] allows 8 possible downslope flow directions, both Cartesian and diagonal directions, to the flow routing scheme. The introduction of triangular facets in the digital terrain analysis allowed for any flow direction between 0° and 360° ($D\infty$) as the steepest downslope direction [Tarboton, 1997]. After the $MD8$ and $D\infty$ algorithms were combined, the $MD\infty$ algorithm was suggested to generate multidirectional flow in any downslope direction [Seibert and McGlynn, 2007]. In this study, the multiple

flow direction algorithm, $MD8$, by Quinn et al. [1991], and the triangular multiple flow direction algorithm, $MD\infty$, by Seibert and McGlynn [2007], were used to compute the spatial distribution of the contributing area and the topographic wetness index. Figure 2 shows the cumulative frequency distributions of the contributing area calculated with $MD8$ and $MD\infty$ using surface and bedrock topography with the monitoring locations and the corresponding contributing areas.

[9] The feasibility of installing the monitoring system and its representation of moisture flow patterns was based on flow divergence for a convex hillslope from $MD8$ and more realistic dispersion for a concave hillslope from the $MD\infty$ contributing areas. The 20 selected monitoring points are shown in Figure 1. Points A to K (25 sensors) were installed for the November 2003 period and additional monitoring points B' to K' (20 sensors) were extended for the May 2004 period. Depending upon the depth of the soil layer, 2 to 3 sensors were installed at the monitoring locations to configure the vertical profile and subsequent redistribution of the soil moisture. Waveguides were inserted parallel to the slope heading in an upslope direction at depths of 10, 30, and 60 cm without disturbing the soil layer. The length of the waveguide cables ranged from 10 m to 50 m depending upon the distance between the monitoring points and the TRASE TDR system.

[10] Temporal variations in soil moisture were recorded hourly during the two periods, which are not rainy seasons in Korea. Rainfall was measured by an automatic rain gauge system (ARGS, Eijkelkamp), located 50 m away from the study area (see Figure 1). Rainfalls of 28 mm, 19 mm, and 8 mm were recorded on 8, 11, and 20 November 2003, while rainfalls of 14.5 mm, 33 mm, and 15.8 mm were recorded for 3, 9, and 20 May 2004, respectively. The 14 and 30 days' antecedent precipitation for the autumn period were 20 and 45.5 mm, and those for the spring period were 47 and 85 mm, respectively. The average rates of potential evapotranspiration (PET) for the autumn and spring periods were estimated to be 1.35 mm/d and 4.9 mm/d, respectively, using meteorological data such as net radiation (pyranometer), wind velocity (cup anemometer), air temperature (platinum resistance sensor), and relative humidity (capacity polymer chip) in 10-min intervals. Leaf area indices (LAIs) were measured using an LAI-2000 [LI-COR Biosciences, 2004] for the corresponding seasonal periods. The LAIs for 5 and 19 November were 1.06 and 0.98, while those for 15 and 28 May were 2.98 and 3.97, respectively.

3. Monitoring Results and Analysis

3.1. Classification of Response Patterns

[11] The soil moisture monitoring results indicate that soil moisture response patterns can be classified into three different zones depending upon the topographic features highlighted by Kim et al. [2007]. The upper zone, located at the top of the hillside, has a relatively low topographic wetness index. Figures 3a and 3b show the soil moisture time series for point A, a monitoring point in the upper zone, during autumn and spring, respectively. Significant increases in soil moisture during rainfall with immediate recessions after rainfall can be observed.

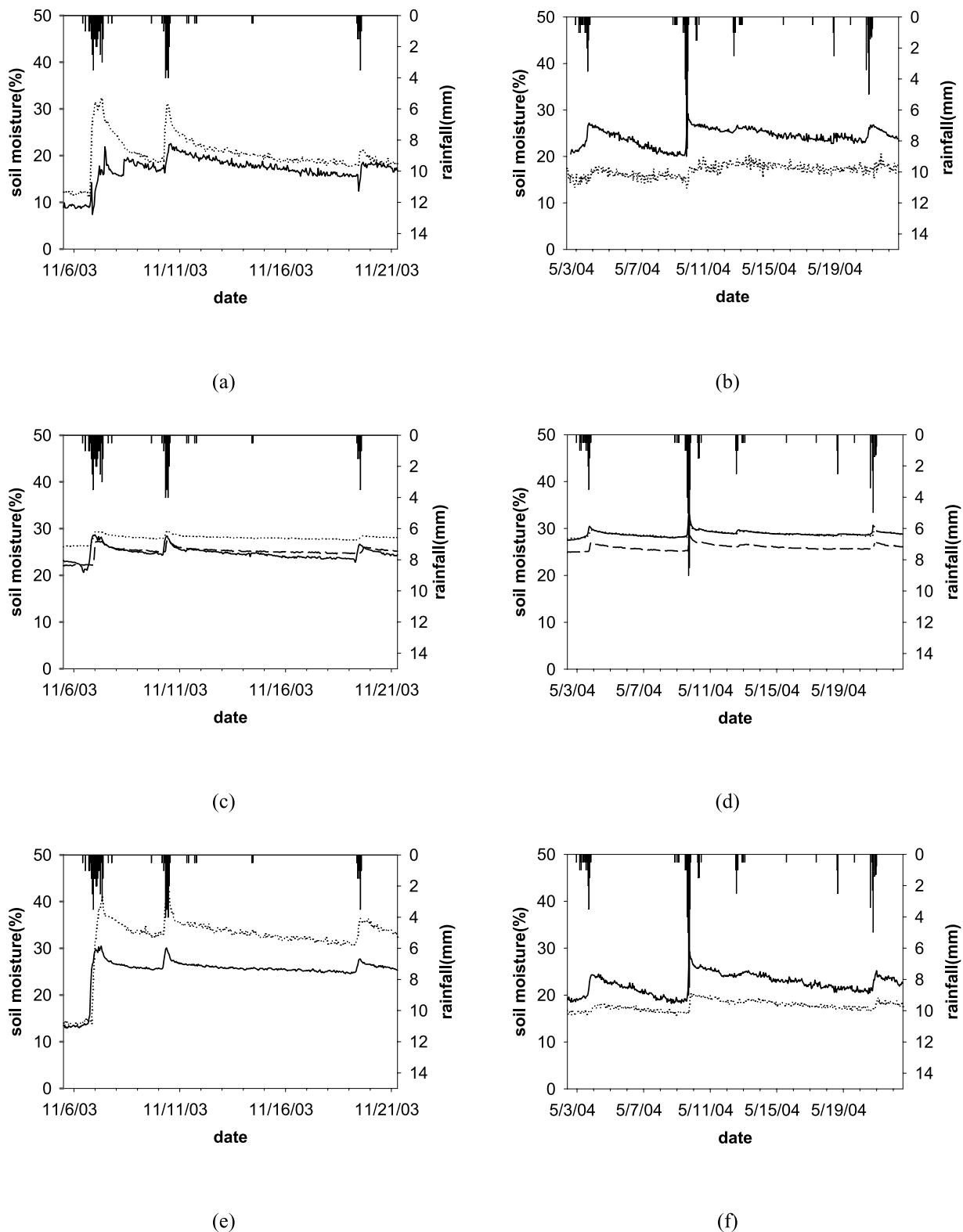


Figure 3. Time series of soil moisture at points (a and b) A in the upper zone, (c and d) G in the middle zone, and (e and f) K in the storage zone for autumn 2003 and spring 2004. Solid line, 10 cm; dotted line, 30 cm; dashed line, 60 cm. The spring data are from Kim *et al.* [2007].

[12] The middle zone has a greater contributing area than the upper zone. Yet the flow tends to diverge and not concentrate, so saturation does not occur. Soil moisture at point G in the middle zone shows a more gradual variation

pattern than the points in the upper zone, as shown in Figures 3c and 3d. A stable variation pattern of the middle zone soil moisture may be associated with the balance

Table 1a. Soil Moisture Statistics for Points A to K^a

		Point										
	Depth (cm)	A	B	C	D	E	F	G	H	I	J	K
6–22 November 2003												
$\bar{X}^b$	10	20.2	23.8	21.4	24.2	23.2	18.3	24.7	19.6	22.0	24.3	25.0
	30	16.9	18.2	25.0	25.8	30.0	25.7	28.0	20.1	20.6	20.3	32.0
	60	-	-	24.6	-	29.7	-	25.1	-	-	-	-
CV ^c	10	0.19	0.12	0.13	0.12	0.09	0.08	0.05	0.18	0.05	0.11	0.14
	30	0.17	0.18	0.07	0.13	0.03	0.04	0.02	0.18	0.08	0.10	0.19
	60	-	-	0.07	-	0.03	-	0.04	-	-	-	-
2–22 May 2004												
$\bar{X}^b$	10	24.1	24.6	21.5	25.1	27.8	25.9	28.8	26.7	11.3	20.3	22.3
	30	17.0	18.3	26.3	20.7	26.9	8.2	28.8	16.1	24.7	18.1	17.7
	60	-	-	20.0	-	31.5	-	25.9	-	-	-	-
CV ^c	10	0.08	0.06	0.06	0.04	0.02	0.03	0.02	0.02	0.07	0.09	0.10
	30	0.08	0.08	0.02	0.04	0.02	0.05	0.02	0.04	0.06	0.12	0.06
	60	-	-	0.03	-	0.02	-	0.02	-	-	-	-

^aNo value means point was below the bedrock.^b $\bar{X}$ indicates the mean of soil moisture. Units are given in percent.^cCV indicates the coefficient of variation.

between the inflow from the upper and upslope zone and the outflow to the lower and downslope directions.

[13] The storage zone has a concave topographic feature with a greater upslope area than the middle zone. A storage zone can often generate overland flow during rainfall events. Figures 3e and 3f represent soil moisture at point K, which is located in the storage zone (see Figure 1). Significant recharge is a dominant characteristic of soil moisture for the storage zone during autumn [Kim *et al.*, 2007]. Slower recessions were observed after rainfall events for soil moisture in the autumn period. Continuous inflow from the upslope area was expected to support persistent soil moisture. Recession is more pronounced for soil moisture in the spring period, as shown in Figure 3f. The higher PET of the spring period explains the steeper recessions during the autumn period, as shown in Figures 3a, 3c, and 3e.

[14] Tables 1a and 1b present the soil moisture time series statistics during 6–22 November 2003 and 2–22 May 2004, respectively. The average autumn soil moisture at the depths of 10 cm and 30 cm for 11 points were 22.4% and 23.8%, respectively, while 22.1% and 19.9% were the corresponding spring means for 20 points. The soil moisture in the autumn showed higher coefficients of variation (CV) than those of the spring. For the autumn data, points A, B, C, and D, which are located near the top of the hillslope, showed higher CVs than those for other lower points such as E, F, G, and I. Points H, J, and K also revealed relatively higher CVs. Therefore, points A, B, C, and D can be classified as the upper zone and points E, F, G, and I as the middle zone, while points H, J, and K are located in the storage zone. The CV for the spring soil moisture shows a spatial distribution similar to that of the autumn, although at a lowered scale. Points A and B have higher CVs than those of points C to I. The CV of point H in the autumn significantly decreased in the spring data. The extended monitoring points B' to K' were stable, except point G' (see Table 1b). Therefore, the extent of the upper zone for autumn is reduced down to points A and B in the spring. Points J, K, and G' are classified as the storage zone, but

point H is excluded during the autumn. Figures 4a and 4b present the spatial distribution of the logarithm for the bedrock contributing area (m²) computed by MD8 and MD∞ with the autumn and spring classifications of the soil moisture response patterns, respectively.

[15] Figure 5 presents the relationship between the CV of the soil moisture and the contributing areas computed by MD∞ using surface and bedrock topography. Figure 5 appears to show that the soil moisture CV is relatively small in the contributing areas between 40 m² and 170 or 200 m² depending upon the surface or bedrock DEM (except in point H). Point H can be excluded in this relationship because the contributing area of point H may not be accurately estimated by MD∞, as the location is highly variable in surface and bedrock topography. This may be associated with ignorance of possible single triangle facets with directions outside the range between 0 and $\pi/4$ located upslope of point H [Seibert and McGlynn, 2007].

3.2. Spatial Patterns

[16] The relationship between the spatial distribution of soil moisture and the topographic index is often studied [Western and Grayson, 1998; Guntner *et al.*, 2004; Sorensen *et al.*, 2006]. Improvement in soil moisture predictability

Table 1b. Soil Moisture Statistics for Points B' to K' for 2–22 May 2004^a

	Depth (cm)	Point								
		B'	C'	C''	D'	E'	G'	F'	I'	K'
$\bar{X}^b$	10	26.9	13.6	23.8	15.5	20.7	16.3	23.2	21.8	20.6
	30	22.6	9.7	16.6	11.2	18.9	23.5	22.7	21.2	27.8
	60	21.9	-	-	-	24.4	-	-	-	-
CV ^c	10	0.04	0.07	0.03	0.09	0.07	0.14	0.04	0.06	0.04
	30	0.03	0.07	0.04	0.08	0.05	0.10	0.03	0.04	0.02
	60	0.03	-	-	-	0.03	-	-	-	-

^aNo value means point was below the bedrock.^b $\bar{X}$ indicates the mean of soil moisture. Units are given in percent.^cCV indicates the coefficient of variation.

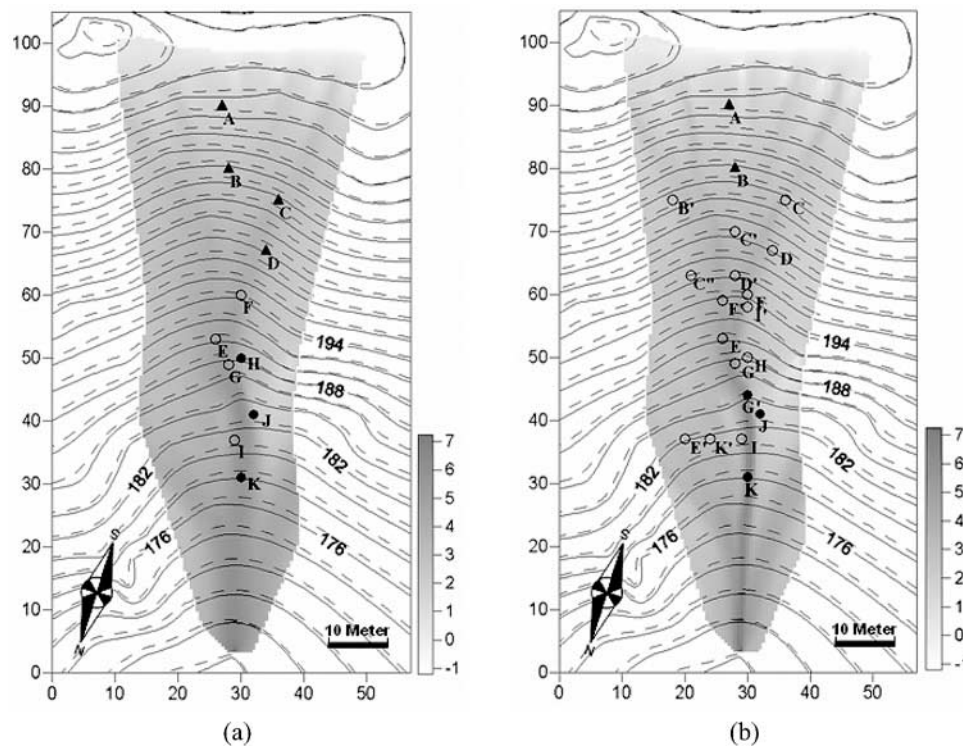


Figure 4. Spatial distribution of $\ln(a)$ using bedrock topography computed by (a) $MD8$ with the autumn classification and (b) $MD\infty$ with the spring classification. Units for a are m^2 and the scale of x and y axes is m; triangles are for the upper zone, open circles are for the middle zone, and the solid circles are for the storage zone; the solid and dashed lines are the surface and bedrock elevation contours, respectively.

can be obtained using bedrock topography such as the DEM [Chaplot and Walter, 2003; Freer *et al.*, 2002]. However, the evaluation of the soil moisture prediction potential has never been performed using spatial distributions of soil moisture means during sequential rainfall-runoff events across two different seasons. Figure 6 shows the soil moisture means of the time series and topographic attributes. The soil moisture is averaged for the two data collection periods and the installed depths. The relationships between the soil moisture means and the topographic surrogates appear to be different depending on the study periods and computational algorithms. Table 2a shows the coefficients of determination from regression analysis between the soil moisture means of the total periods and the estimated soil moisture according to terrain surrogates (e.g., contributing areas or the topographic index) for data collection conditions using regression models such as linear, logarithmic, exponential, and power law equations. For autumn, the data set for the 10-cm depth shows a weak correlation while a substantial relationship can be found for the data set for the 30-cm depth. Close inspection of Table 2a reveals the following: (1) digital terrain analysis using bedrock topography improves prediction accuracy; (2) the prediction capability of $MD\infty$ is higher than that of $MD8$; and (3) regression analyses for the contributing areas using logarithm and power law equations provide better explanations of the soil moisture distribution than the linear and exponential equations. Table 2a also indicates that a negligible correlation exists between the soil moisture during the spring and the terrain analysis. In order to further explore the relationship between the topography and the subsets of

soil moisture histories, the soil moisture means for three subsections, representing each rainfall event, were estimated for both periods and used for regression analysis. Apparent similarities were found in the coefficients of determination of the correlations between the soil moisture subsets with shorter time periods as well as between those for the means of the total soil moisture and the means of the shorter subsets (see Tables 2a, 2b, 2c, and 2d). The role of topography in determining soil moisture is apparently different in the two seasons.

3.3. Lag Times

[17] Soil moisture variation in response to rainfall input can be described as a peak connected with a rising limb and a recession limb. The soil moisture response pattern at a point scale can be determined by various factors, such as the infiltration structure of a soil matrix, the antecedent moisture content, and the rainfall rate. In this paper, the lag time, defined as the difference between the time of the maximum hourly rainfall rate and that of the peak soil moisture, is used to characterize the temporal response of soil moisture between the two seasons. Tables 3a and 3b present the lag times of the monitoring points for autumn and spring, respectively, with the antecedent moisture content and the corresponding rainfall events. The mean antecedent moisture content for autumn and spring was 19.7% and 22.0% at a depth of 10 cm, and 21.1% and 19.1% at a depth of 30 cm, respectively. Tables 3a and 3b indicate that higher rainfall rates caused faster responses within each seasonal period. Generally, longer rainfall duration and more dispersed temporal distribution result in the longer lag time. However,

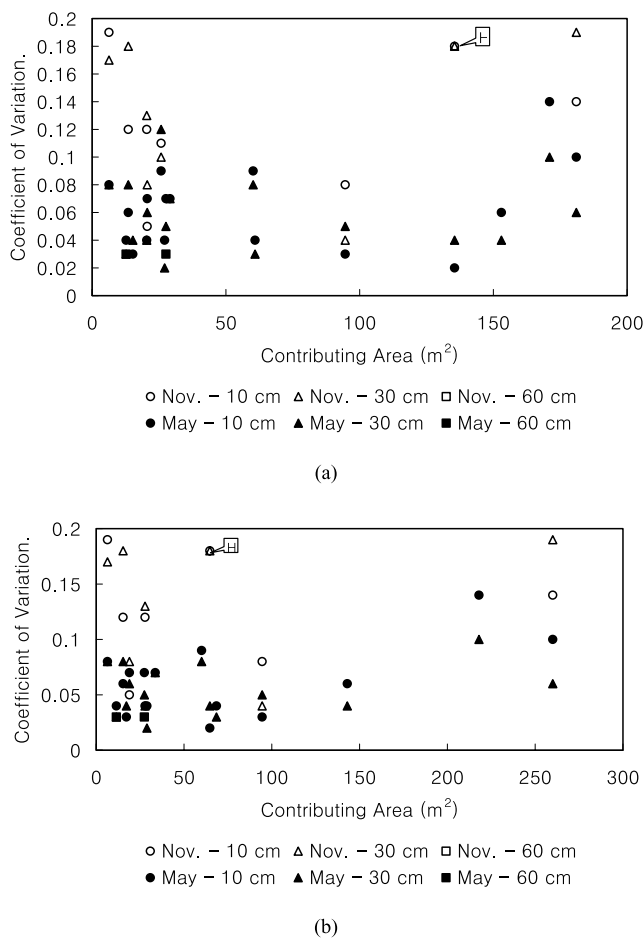


Figure 5. Coefficient of variation for soil moisture and contributing area computed by (a) MD_{∞} with surface topography and (b) MD_{∞} with bedrock topography; the H in the box is for point H.

the lag time scales of the spring events, 2 and 9 May, were larger than those for autumn, 8 and 11 November even with the similar sequential rainfall durations between two seasons (Tables 3a and 3b) and more concentrated temporal distribution during the spring period (as shown in Figure 8). The soil moisture transfer process in the autumn showed prompt responses for a maximum rainfall rate of 4 mm/h, but the soil moisture during the spring for a rainfall rate of 9 mm/h showed slower lag times in similar duration and antecedent moisture conditions (Tables 3a and 3b).

3.4. Storage Distributions

[18] Temporal and spatial distribution of the soil water content on the hillslope is an important factor in understanding hydrological processes such as water storage and runoff generation. Water storage history can be estimated from soil moisture measurements by adding the multiplication of a soil moisture time series and its corresponding soil depth intervals. However, measured soil moisture can vary depending on vertical infiltration, percolation to a deeper layer, lateral upslope contribution, lateral downslope outflow, and evapotranspiration back to the atmosphere [Ridolfi *et al.*, 2003; Kim and Kim, 2007]. The relationship between cumulative moisture storage and cumulative rainfall can

reveal the response of water to vertical rainfall input and redistribution on the point scale. Estimated maximum canopy storages using measured LAIs range from 0.212 mm/d to 0.196 mm/d in the autumn period and between 0.894 mm/d and 0.600 mm/d in the spring period [Dickinson, 1984]. Considering the scales of the rainfall events and durations in Tables 3a and 3b, the amount of water intercepted by the forest canopy could be ignored. In this approach, rainfall data from an open area was used. Figure 7 shows the relationship between the cumulative storage and the cumulative rainfall for points A, C, E, G, J, and K during November 2003 and May 2004. Water storage during the autumn for points A and C varied positively with a significant variation. However, water storage for the spring showed a partially negative response (i.e., water loss), as shown in Figures 7a and 7b. The scale of the water storage variation for point A was reduced in point C. Figures 7c and 7d show that the water storage variation for points E and G is more stable than that of other points. Negative water storage for upper points disappear in Figures 7c and 7d. Lateral flow from an upslope area and reduced evapotranspiration may be possible explanations for the positive water storage at points E and G during the autumn. The difference associated with the seasonal patterns seems minor at points E and G, while the differences in the cumulative storage for rainfall events between the two seasons are 6.1 mm and 3 mm for points E and G, respectively. Water storage responses at points J and K are shown in Figures 7e and 7f. Seasonal differences appeared to be most pronounced at point K. A significant amount of recharge was observed from the storage history of the autumn period, whereas a negligible variation in water storage was observed for the spring period. The cumulative storage for points J and K, the storage zone, at the end of May 2004 was 6 mm and 5 mm, respectively, while equivalent values at the end of November 2003 were 29.2 mm and 60.2 mm, respectively.

[19] Given the variation in water storage, water storage response patterns can be categorized into three classes: (1) the upper zone, which exhibits immediate recharge and persistent recession (points A and C (excluding point C for May 2004)); (2) the middle zone, which exhibits a stable response (points E and G for November 2003 and points C, E, and G for May 2004); and (3) the storage zone, which exhibits storage increments due to upslope lateral flow and points J and K. The water storage characteristics of the other points can also be classified into one of the above described response patterns.

3.5. Multiple Regression Recharge

[20] Infiltration and upslope lateral flow during rainfall events increase soil moisture, while vertical percolation to deeper soil, downslope lateral outflow, and evapotranspiration decrease soil moisture [Ridolfi *et al.*, 2003; Kim and Kim, 2007]. In this study, soil moisture recharge is defined as the difference between the antecedent moisture content and the peak soil moisture. A multiple regression analysis was conducted for soil moisture recharge using the antecedent moisture content and total rainfall amounts as the dependent variables. Table 4 shows several statistics obtained from multiple regression analysis for soil moisture recharge at different depths and periods. Table 4 reveals that more accurate predictions, both in terms of the multiple regression coefficient (MRC) and the adjusted coefficient of

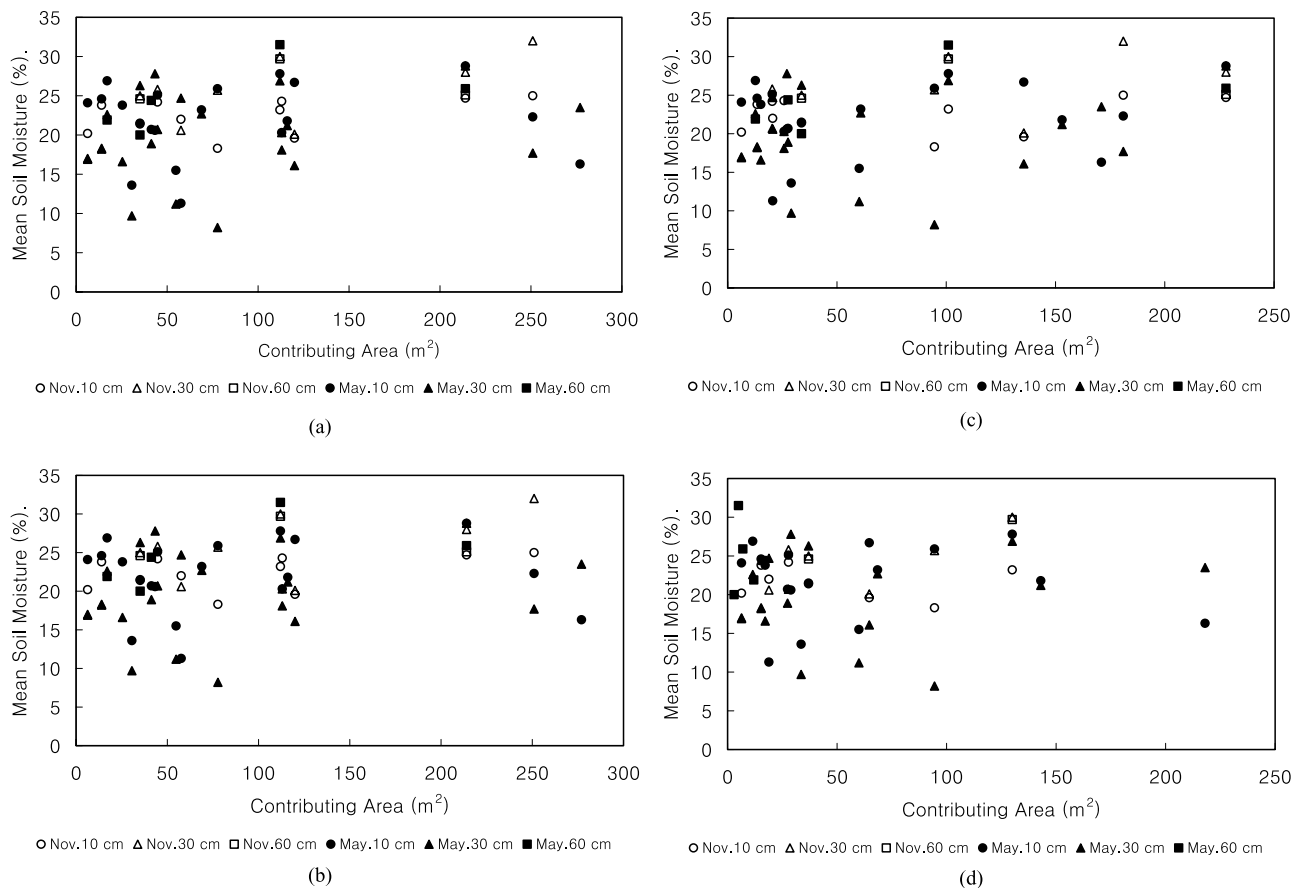


Figure 6. Soil moisture and contributing areas computed by (a) *MD8* with surface topography, (b) *MD8* with bedrock, (c) *MD ∞* with surface topography, and (d) *MD ∞* with bedrock topography and soil moisture means and the topographic index computed by (e) *MD8* with surface topography, (f) *MD8* with bedrock, (g) *MD ∞* with surface topography, and (h) *MD ∞* with bedrock topography.

determination (AR2), can be found for the 10-cm depth than for the 30-cm depth. Infiltration of water to a deeper depth will naturally face more obstacles (i.e., macropore or water uptake by root) in reaching the sensor than at a shallow depth. The MRCs and AR2s during the autumn period were higher than those for spring in the upper and middle zones. The storage zone showed similar MRCs and AR2s for the spring and the autumn periods. The standard errors (SE) for the regression analysis during the spring period were lower than in the autumn except for the middle zone. Fitness to normal distribution, R2PN, in Table 4, to predictions is generally high in the spring period. Distinctions in the multiple regression coefficient, adjusted coefficient of determination, and standard error between the middle zone and the storage zone seem apparent in comparative seasonal regression analysis.

3.6. Runoff

[21] Runoff time series were obtained from the water level gauge, as shown in Figure 1. The distance between the streamflow gauging station and the hillslope is 1,230 m. The generation of streamflow is exclusively associated with the rainfall event because there is no flow control structure between the water level gauge and the soil moisture measurement site. Figures 8a and 8b show the rainfalls and runoff

responses during the soil moisture monitoring periods in Figure 3. The initial streamflow rates for 8 November 2003 and 3 May 2004 were 0.028 and 0.142 m³/s, respectively, while the peak runoff rates for the corresponding events were estimated to be 0.188 and 0.241 m³/s, respectively. The lag times between the rainfall peaks and the runoff peaks were 28, 27, and 31 h for the rainfall events on 8, 11, and 20 November 2003, respectively, whereas those from the spring were 4, 3, and 4 h for the rainfall events on 3, 9, and 20 May 2004, respectively. The difference in the runoff response patterns between the two seasons is apparent, even with similar rainfall distribution and mean soil moisture values. Seasonal differences in initial streamflow and distinct soil moisture response patterns are primarily associated with differences in runoff responses.

4. Discussion

4.1. Seasonal Difference and Ecohydrological Processes

[22] The average rainfall rates were estimated to be 3.9 and 3.7 mm/d, while the soil moisture means were 23.1% and 21.0% for the autumn and spring periods, respectively. However, the soil moisture response characteristics were significantly different. During the autumn, the coefficients of variation were greater than those for the spring. The

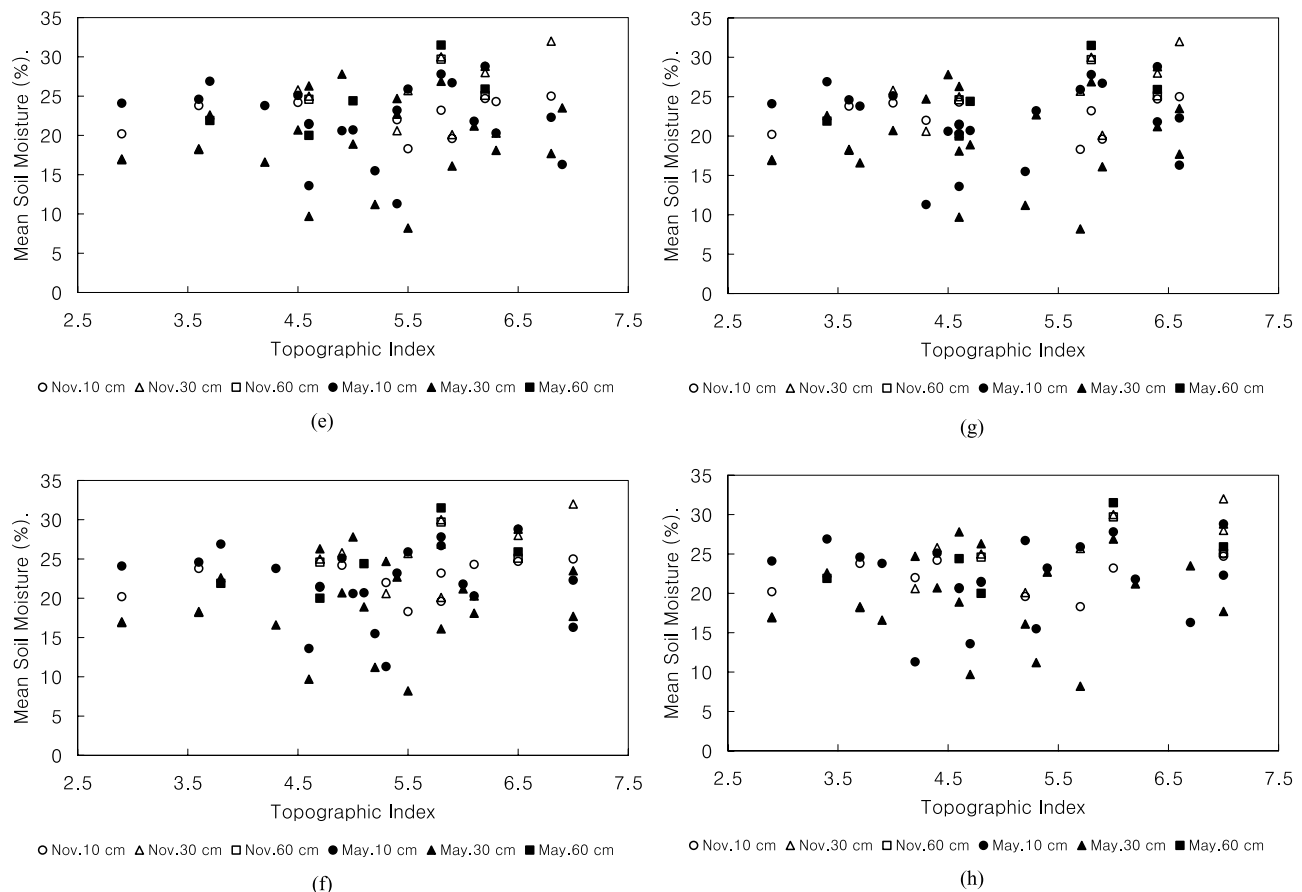


Figure 6. (continued)

predictability of soil moisture recharge based on the antecedent moisture content and precipitation for the autumn period is also higher than that for the spring period in the upper and middle zones. The increasing trend during the autumn period and steady fluctuation behavior for the spring period are primarily associated with the greater storage recharge in November rather than that for May.

[23] The influence of vegetation, canopy cover, and litter density is higher in the spring than in the autumn [Baird and Wilby, 1999; Rodriguez-Iturbe and Porporato, 2004]. The estimated PET rate and the maximum canopy storage for November were less than one third of those during May. The higher evapotranspiration and greater interception losses could explain the longer lag times and faster recessions of soil moisture (Tables 3a and 3b) during the spring period. More loss from PET and substantial canopy interception during the spring also explained the seasonal difference in the soil moisture recharges shown in Figure 7. However, the shorter lag times (Tables 3a and 3b) during the autumn, especially those of points D to K, in the middle and storage zones, cannot be completely explained by seasonal differences in interception and PET even with a similarity in rainfall duration and more concentrated distribution in rainfall intensity during the spring period, as shown in Figure 8. The scales of the maximum or mean rainfall rates were significantly greater than those of interception and PET, while the time scale for the peak soil moisture was short for both periods, as shown in Tables 3a and 3b. The seasonal difference in soil moisture recharge predictability

(Table 4) also seems to be responsible for the seasonal differences in the infiltration rate, as well as interception and PET during the rainfall events. Living and decayed roots contributed to preferential flow pathways, while the conductivities of a macropore increase as roots decay [Kozłowski *et al.*, 1997; Noguchi *et al.*, 1999; Das Gupta *et al.*, 2006]. Seasonal differences in the measured LAI values also indicate significant differences in fine root density [Gower *et al.*, 1992; Haggard and Ewel, 1995; Tingey *et al.*, 1996]. Fine root decomposition before the autumn period and subsurface erosion with pipe flow may improve the connectivity for pipe flow [Selby, 1993; Noguchi *et al.*, 1997; Uchida *et al.*, 1999].

[24] Considering the negligible difference of mean soil moisture and rainfall distributions, and the significant difference in the topographic control to soil moisture between the two seasons, the difference in the spatial organization of soil moisture between the wet and dry periods appears not to be applicable in the study area [Western and Grayson, 1998]. The impacts of surface and subsurface vegetation should be further considered to represent the ecohydrological dynamics of soil moisture [Rodriguez-Iturbe and Porporato, 2004; Tromp-van Meerveld and McDonnell, 2006].

4.2. Topographic Control and Soil Moisture Characterization

[25] Depending on the degree of saturation, there are numerous control factors for soil moisture. In dry conditions, the water retention characteristics of the soil and

Table 2a. Coefficients of Determination, r^2 , for Both Seasons From Total Periods^a

Regression Equations	6–22 Nov 2003		2–22 May 2004	
	10 cm	30 cm	10 cm	30 cm
<i>MD8 Surface</i>				
Linear	0.13 (0.08)	0.35 (0.38)	0.00 (0.01)	0.03 (0.02)
Logarithm	0.07 (0.07)	0.40 (0.39)	0.00 (0.02)	0.03 (0.02)
Exponential	0.12 (0.07)	0.34 (0.41)	0.00 (0.02)	0.03 (0.02)
Power	0.07 (0.14)	0.42 (0.39)	0.00 (0.04)	0.02 (0.02)
<i>MD8 Bedrock</i>				
Linear	0.20 (0.10)	0.49 (0.48)	0.00 (0.01)	0.05 (0.03)
Logarithm	0.11 (0.11)	0.50 (0.50)	0.00 (0.00)	0.04 (0.04)
Exponential	0.18 (0.11)	0.45 (0.55)	0.00 (0.01)	0.05 (0.02)
Power	0.09 (0.20)	0.52 (0.50)	0.00 (0.03)	0.03 (0.04)
<i>MD∞ Surface</i>				
Linear	0.04 (0.02)	0.43 (0.57)	0.05 (0.00)	0.03 (0.01)
Logarithm	0.01 (0.02)	0.55 (0.56)	0.02 (0.00)	0.01 (0.01)
Exponential	0.03 (0.10)	0.42 (0.58)	0.05 (0.03)	0.02 (0.00)
Power	0.01 (0.03)	0.56 (0.57)	0.01 (0.10)	0.00 (0.02)
<i>MD∞ Bedrock</i>				
Linear	0.24 (0.14)	0.53 (0.76)	0.04 (0.00)	0.07 (0.02)
Logarithm	0.11 (0.09)	0.73 (0.75)	0.01 (0.00)	0.02 (0.02)
Exponential	0.22 (0.10)	0.50 (0.75)	0.03 (0.00)	0.05 (0.02)
Power	0.09 (0.23)	0.73 (0.74)	0.01 (0.00)	0.01 (0.05)

^aHere r^2 represents explained variation divided by total variation between averages of measured and estimated soil moisture using contributing areas for several regression analysis conditions such as computing algorithm, used topography, date, soil depth, and regression models; Linear equation is $y = ax + b$, logarithm equation is $y = a \ln(x) + b$, exponential equation is $y = a \exp(bx)$, and power equation is $y = ax^b$, where y and x are mean soil moistures and modeled contributing areas and a and b are constants for regression models, respectively. Numbers in parentheses are r^2 when x is the topographic index.

Table 2b. Coefficients of Determination, r^2 , for the First Subset of Data From Total Periods^a

Regression Equations	6–10 Nov 2003		2–8 May 2004	
	10 cm	30 cm	10 cm	30 cm
<i>MD8 Surface</i>				
Linear	0.07 (0.05)	0.25 (0.38)	0.04 (0.00)	0.03 (0.03)
Logarithm	0.04 (0.04)	0.40 (0.41)	0.00 (0.00)	0.03 (0.03)
Exponential	0.05 (0.04)	0.25 (0.41)	0.04 (0.04)	0.03 (0.02)
Power	0.03 (0.03)	0.43 (0.45)	0.00 (0.03)	0.02 (0.02)
<i>MD8 Bedrock</i>				
Linear	0.15 (0.09)	0.35 (0.48)	0.00 (0.01)	0.05 (0.03)
Logarithm	0.07 (0.07)	0.48 (0.49)	0.00 (0.00)	0.04 (0.04)
Exponential	0.11 (0.07)	0.34 (0.51)	0.00 (0.01)	0.05 (0.02)
Power	0.05 (0.05)	0.51 (0.53)	0.00 (0.03)	0.03 (0.04)
<i>MD∞ Surface</i>				
Linear	0.04 (0.01)	0.34 (0.50)	0.14 (0.02)	0.04 (0.01)
Logarithm	0.01 (0.01)	0.47 (0.51)	0.05 (0.01)	0.02 (0.01)
Exponential	0.02 (0.00)	0.31 (0.49)	0.12 (0.02)	0.02 (0.00)
Power	0.00 (0.00)	0.46 (0.52)	0.04 (0.00)	0.00 (0.00)
<i>MD∞ Bedrock</i>				
Linear	0.31 (0.13)	0.43 (0.66)	0.11 (0.02)	0.08 (0.03)
Logarithm	0.12 (0.09)	0.63 (0.68)	0.04 (0.00)	0.03 (0.03)
Exponential	0.27 (0.10)	0.39 (0.63)	0.10 (0.02)	0.05 (0.01)
Power	0.09 (0.07)	0.60 (0.67)	0.04 (0.01)	0.01 (0.01)

^aHere r^2 represents explained variation divided by total variation between averages of measured and estimated soil moisture using contributing areas for several regression analysis conditions such as computing algorithm, used topography, date, soil depth, and regression models; Linear equation is $y = ax + b$, logarithm equation is $y = a \ln(x) + b$, exponential equation is $y = a \exp(bx)$, and power equation is $y = ax^b$, where y and x are mean soil moistures and modeled contributing areas and a and b are constants for regression models, respectively. Numbers in parentheses are r^2 when x is the topographic index.

Table 2c. Coefficients of Determination, r^2 , for the Second Subset of Data From Total Periods^a

Regression Equations	10–15 Nov 2003		8–15 May 2004	
	10 cm	30 cm	10 cm	30 cm
<i>MD8 Surface</i>				
Linear	0.12 (0.00)	0.38 (0.38)	0.05 (0.00)	0.03 (0.02)
Logarithm	0.03 (0.02)	0.38 (0.37)	0.00 (0.00)	0.03 (0.02)
Exponential	0.03 (0.00)	0.42 (0.56)	0.11 (0.01)	0.01 (0.00)
Power	0.03 (0.18)	0.40 (0.40)	0.00 (0.00)	0.02 (0.01)
<i>MD8 Bedrock</i>				
Linear	0.16 (0.06)	0.51 (0.48)	0.06 (0.00)	0.04 (0.03)
Logarithm	0.15 (0.04)	0.47 (0.46)	0.01 (0.00)	0.04 (0.03)
Exponential	0.15 (0.05)	0.49 (0.50)	0.06 (0.00)	0.04 (0.02)
Power	0.04 (0.05)	0.50 (0.50)	0.01 (0.00)	0.03 (0.02)
<i>MD∞ Surface</i>				
Linear	0.03 (0.01)	0.43 (0.55)	0.03 (0.01)	0.43 (0.55)
Logarithm	0.00 (0.00)	0.51 (0.53)	0.00 (0.00)	0.51 (0.53)
Exponential	0.03 (0.00)	0.42 (0.56)	0.03 (0.00)	0.42 (0.56)
Power	0.00 (0.00)	0.53 (0.55)	0.00 (0.00)	0.53 (0.55)
<i>MD∞ Bedrock</i>				
Linear	0.20 (0.07)	0.53 (0.73)	0.11 (0.02)	0.06 (0.02)
Logarithm	0.07 (0.05)	0.69 (0.71)	0.04 (0.01)	0.02 (0.02)
Exponential	0.18 (0.06)	0.51 (0.74)	0.10 (0.02)	0.04 (0.01)
Power	0.06 (0.04)	0.70 (0.74)	0.04 (0.01)	0.01 (0.00)

^aHere r^2 represents explained variation divided by total variation between averages of measured and estimated soil moisture using contributing areas for several regression analysis conditions such as computing algorithm, used topography, date, soil depth, and regression models; Linear equation is $y = ax + b$, logarithm equation is $y = a \ln(x) + b$, exponential equation is $y = a \exp(bx)$, and power equation is $y = ax^b$, where y and x are mean soil moistures and modeled contributing areas and a and b are constants for regression models, respectively. Numbers in parentheses r^2 when x is the topographic index.

Table 2d. Coefficients of Determination, r^2 , for the Third Subset of Data From Total Periods^a

Regression Equations	15–22 Nov 2003		15–22 May 2004	
	10 cm	30 cm	10 cm	30 cm
<i>MD8 Surface</i>				
Linear	0.18 (0.10)	0.37 (0.41)	0.04 (0.00)	0.03 (0.02)
Logarithm	0.10 (0.09)	0.42 (0.42)	0.00 (0.01)	0.02 (0.02)
Exponential	0.17 (0.10)	0.35 (0.43)	0.04 (0.00)	0.03 (0.01)
Power	0.10 (0.09)	0.45 (0.45)	0.00 (0.01)	0.01 (0.01)
<i>MD8 Bedrock</i>				
Linear	0.23 (0.14)	0.50 (0.51)	0.05 (0.00)	0.04 (0.03)
Logarithm	0.13 (0.12)	0.51 (0.50)	0.01 (0.00)	0.03 (0.02)
Exponential	0.22 (0.14)	0.47 (0.53)	0.05 (0.00)	0.04 (0.02)
Power	0.13 (0.12)	0.54 (0.54)	0.00 (0.00)	0.02 (0.01)
<i>MD∞ Surface</i>				
Linear	0.07 (0.04)	0.44 (0.58)	0.12 (0.00)	0.03 (0.01)
Logarithm	0.04 (0.05)	0.55 (0.58)	0.04 (0.00)	0.01 (0.01)
Exponential	0.06 (0.04)	0.42 (0.59)	0.11 (0.01)	0.02 (0.00)
Power	0.04 (0.04)	0.57 (0.60)	0.03 (0.00)	0.00 (0.00)
<i>MD∞ Bedrock</i>				
Linear	0.25 (0.16)	0.53 (0.77)	0.10 (0.02)	0.06 (0.02)
Logarithm	0.16 (0.14)	0.73 (0.77)	0.03 (0.00)	0.02 (0.01)
Exponential	0.24 (0.15)	0.50 (0.77)	0.09 (0.02)	0.04 (0.01)
Power	0.15 (0.13)	0.74 (0.79)	0.03 (0.01)	0.01 (0.00)

^aHere r^2 represents explained variation divided by total variation between averages of measured and estimated soil moisture using contributing areas for several regression analysis conditions such as computing algorithm, used topography, date, soil depth, and regression models; Linear equation is $y = ax + b$, logarithm equation is $y = a \ln(x) + b$, exponential equation is $y = a \exp(bx)$, and power equation is $y = ax^b$, where y and x are mean soil moistures and modeled contributing areas and a and b are constants for regression models, respectively. Numbers in parentheses r^2 when x is the topographic index.

Table 3a. Lag Times Between the Maximum Rainfall and Soil Moisture Peak for November 2003

Point	10 cm Depth		30 cm Depth		60 cm Depth	
	Lag Time (h)	AM ^a (%)	Lag Time (h)	AM ^a (%)	Lag Time (h)	AM ^a (%)
<i>8 November 2003^b</i>						
A	2	12	6	9		
B	2	15	9	10		
C	2	13	2	20	8	20
D	1	18	1	17		
E	1	17	2	28	3	27
F	2	16	3	23		
G	2	23	2	26	3	22
H	2	8	2	9		
I	1	18	2	19		
J	2	17	4	14		
K	2	14	3	14		
<i>11 November 2003^c</i>						
A	0	19	4	17		
B	0	23	3	16		
C	0	22	0	25	4	25
D	0	23	0	25		
E	0	23	0	30	0	30
F	0	18	1	26		
G	0	25	0	28	0	25
H	0	20	0	21		
I	0	21	0	22		
J	0	25	1	21		
K	0	26	1	34		
<i>20 November 2003^d</i>						
A	5	18	5	16		
B	4	24	8	18		
C	2	19	8	24	15	24
D	0	23	0	25		
E	1	22	0	29	7	29
F	0	17	8	25		
G	1	24	4	28	8	25
H	1	19	4	20		
I	2	20	6	21		
J	1	23	7	19		
K	1	25	11	21		

^aAM, antecedent moisture content.^bRain sum, 28 mm; mean rate, 1.4 mm/h; maximum rate 3 mm/h; duration time, 17 h.^cRain sum, 19 mm; mean rate, 2.7 mm/h; maximum rate 4 mm/h; duration time, 8 h.^dRain sum, 8 mm; mean rate, 1.3 mm/h; maximum rate 2.5 mm/h; duration time, 7 h.

surface cover are dominant components. In addition, soil porosity can be a significant factor in nearly saturated soil layers [Dendrou and Dendrou, 1999; Canton *et al.*, 2004]. Topography is considered to be the most dominant factor at intermediate saturation levels [Anderson and Kneale, 1980; Wilson *et al.*, 2005]. Figure 5 also shows that the spatial distribution of soil moisture can be identified as the upper zone, the middle zone, and the storage zone in terms of the CV of the contributing area. However, the middle zone appeared to expand in the spring, as noted in Tables 1a and 1b. The CVs of points C, D, and H during the autumn were significantly reduced in the spring. The longer lag times shown in Tables 3a and 3b during the spring demonstrate consistent results in middle zone expansion. The similarity in water storage for points E and G is more apparent than for the transitional points C, D, and H, as shown in Figures 4 and 7. The recharge feature of points J and K, and the

storage zone during the autumn, disappeared in the spring. Soil moistures at the middle zone have lower CVs than those of the storage zone. This can be explained by pointing out that the downslope flow boundaries of the middle zone and the storage zone are different. The storage zone can act as the source area for streamflow generation. The higher upslope area of the storage zone is associated with greater recharge impact than that of the middle zone, while the recession scale of the storage zone is also influenced by the downslope boundary condition for streamflow generation.

[26] The regression analysis in Tables 2a–2d indicates that a significant relationship between the mean soil moisture and the topographic surrogates can only be found at a depth of 30 cm in the autumn period. Higher correlations of bedrock topographic attributes and soil moisture reveal that the hydrological process in the study area is related to the soil water movement in the hillslope being governed by

Table 3b. Lag Times Between the Maximum Rainfall and Soil Moisture Peak for May 2004

Point	10 cm Depth		30 cm Depth		60 cm Depth	
	Lag Time (h)	AM ^a (%)	Lag Time (h)	AM ^a (%)	Lag Time (h)	AM ^a (%)
<i>3 May 2004^b</i>						
A	2	21	11	14		
B	4	22	17	16		
C	4	19	14	26	24	20
D	10	24	16	20		
E	1	26	10	26	13	30
F	3	24	11	8		
G	1	28	3	28	4	25
H	3	26	10	15		
I	6	10	4	22		
J	8	17	4	15		
K	14	19	13	16		
<i>9 May 2004^c</i>						
A	2	20	5	14		
B	1	22	11	16		
C	2	19	4	25	23	19
D	4	23	10	19		
E	1	27	3	26	7	31
F	2	25	7	8		
G	1	28	2	28	1	25
H	2	26	3	15		
I	1	10	1	23		
J	1	17	1	15		
K	1	19	4	16		
<i>20 May 2004^d</i>						
A	2	23	7	16		
B	0	24	3	18		
C	2	21	7	26	24	20
D	5	25	5	20		
E	6	28	5	27	7	31
F	2	25	10	8		
G	0	29	2	29	6	26
H	4	26	4	16		
I	2	11	6	25		
J	4	19	5	17		
K	5	21	7	17		

^aAM, antecedent moisture content.^bRain sum, 14.5 mm; mean rate, 0.9 mm/h; maximum rate 3.5 mm/h; duration time, 17 h.^cRain sum, 33 mm; mean rate, 3.7 mm/h; maximum rate 9 mm/h; duration time, 9 h.^dRain sum, 15.8 mm; mean rate, 1.6 mm/h; maximum rate 5 mm/h; duration time, 10 h.

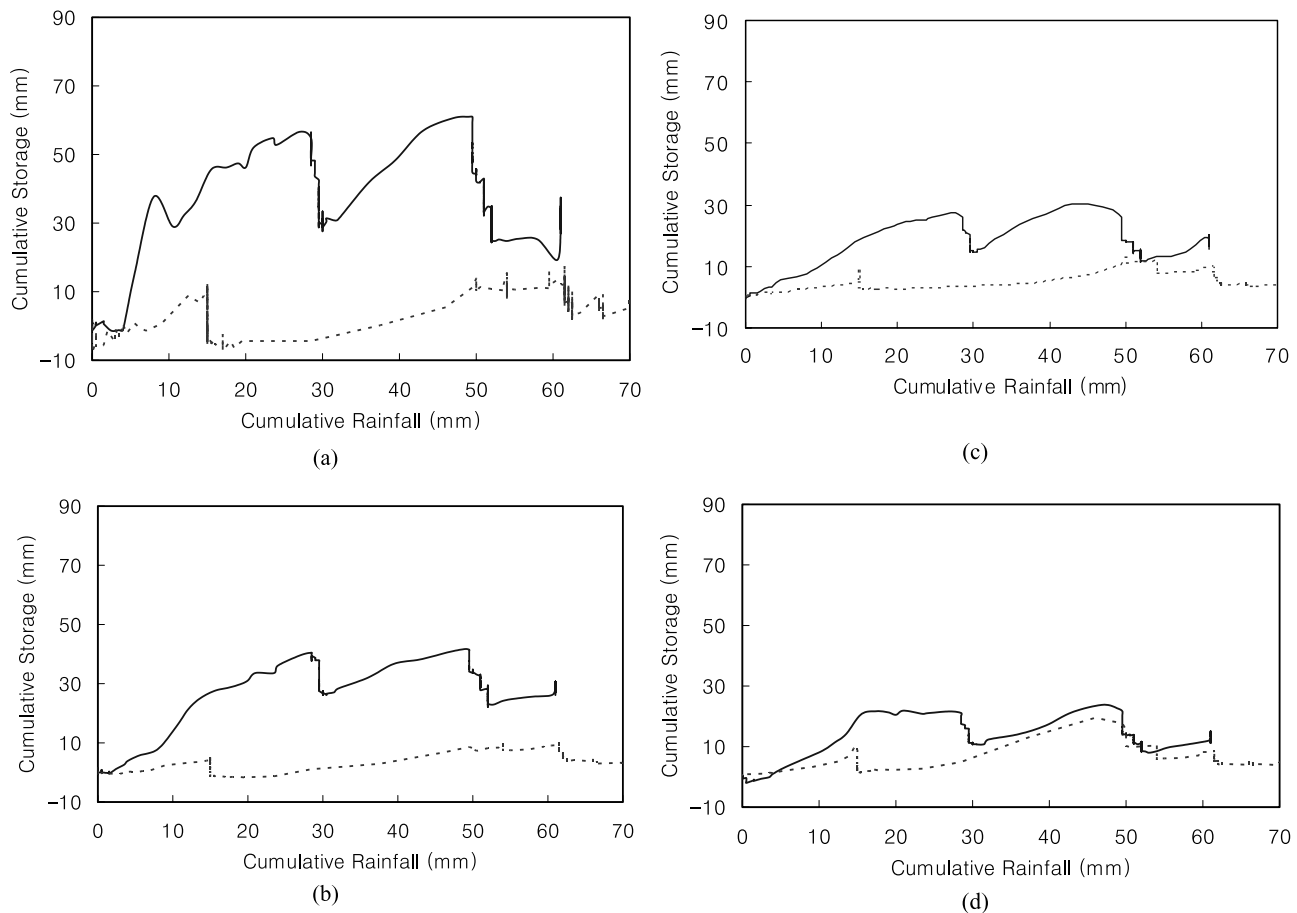


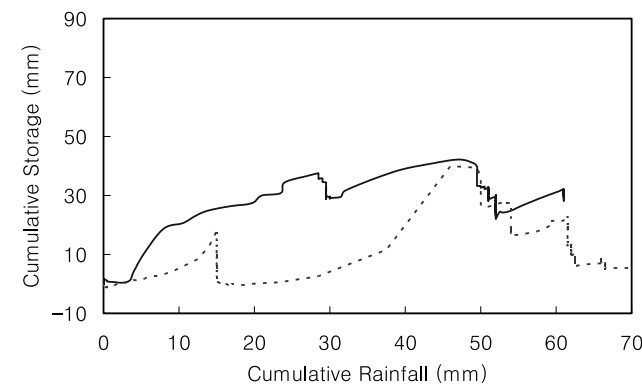
Figure 7. Cumulative water storage and cumulative rainfall for (a) point A, (b) point C, (c) point E, (d) point G, (e) point J, and (f) point K; the solid line is for the autumn period, and the dashed line is for the spring period.

vertical drainage to weathered bedrock, and the subsequent generation of the lateral flow along bedrock [McGlynn and McDonnell, 2003; Uchida *et al.*, 2003b; Haga *et al.*, 2005]. Soil moisture at 60 cm also showed similar results (Tables 1a and 1b), but regression analysis with topography was not performed because of the small number of monitoring points and the shallow average soil depth. Faster lag times (Tables 3a and 3b) and better prediction of soil moisture recharge (Table 4) during the autumn are strongly associated with a higher correlation between the mean soil moisture and the bedrock topographic feature for that period. Significant infiltration in the vertical direction appears to be the necessary condition for lateral redistribution of soil moisture. The impact of flow seems more pronounced, and soil moisture was less controlled by evapotranspiration and vegetation during the autumn period. Lateral flow along the surface seems negligible, and thin saturated flow over bedrock may contribute to the lateral distribution of soil moisture [Buttle and McDonald, 2002] when the scale of the rainfall event and soil moisture increments are considered.

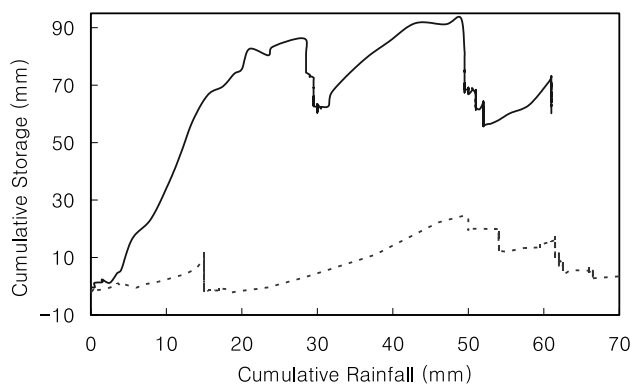
4.3. Runoff Responses and Soil Moisture Characteristics

[27] As can be seen in Figures 8a and 8b, the shorter lag times between the rainfall peaks and the soil moisture peaks

are associated with the longer lag times of the runoff responses, and vice versa for the longer soil moisture lag times. The high correlation between subsurface topography and soil moisture distribution shown in Tables 2a–2d and the rapid vertical flux shown in Tables 3a and 3b indicate that water movement during the autumn was primarily driven by subsurface flow. The significant differences in cumulative water storage (Figure 7) between the two seasons explained the substantial differences in the runoff responses, both in the time scale and runoff volumes. The long runoff lag times in the autumn events can be explained by the expansion of the saturated zone over the upslope area driven by flow dominant control (Tables 2a–2d) and the high water-storing feature of the soil layer (Figure 7), which could be associated with the seasonal difference in the soil hydraulic properties [Das Gupta *et al.*, 2006; Bormann and Klaassen, 2008]. Saturated subsurface flow above the bedrock and dominant vertical flow may lead the connection between the upslope area and the saturated zone over the soil-bedrock interface (Tables 2a–2d and Figure 7) during events with long runoff lag times in the autumn [Buttle and McDonald, 2002; Freer *et al.*, 2002]. On the basis of the catchment size at the stream gauge, 5.03 km², the time scale for base flow separation is estimated to be 27 h by the N day technique [McCuen, 2004]. This indicates that most of the streamflow might originate from the base



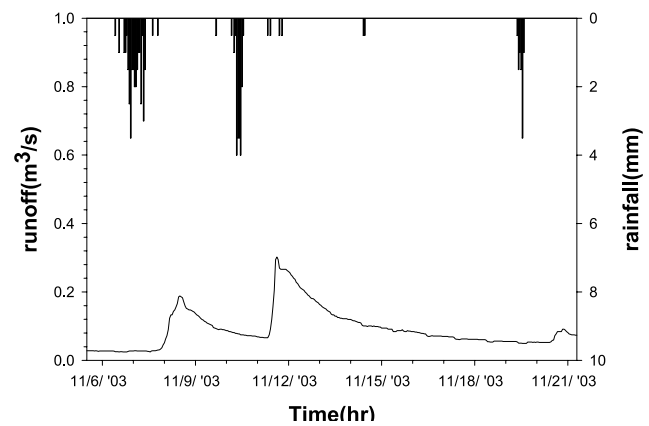
(e)



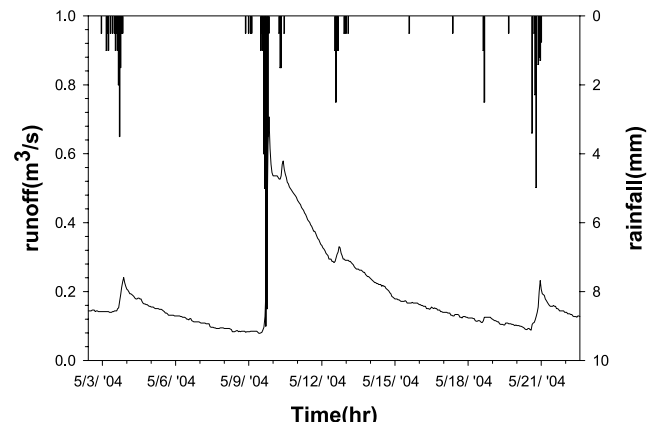
(f)

Figure 7. (continued)

flow during the autumn periods. Considering the vertical distribution of macropores [Beven and Germann, 1982], the matrix flow could be more responsible for the delivery of water in deeper depths than the flow through macropores and results in the long lag times for the runoff response [Pearce *et al.*, 1986]. During the rainfall events in the spring, the source area for runoff generation seemed to be limited to the storage zone, which resulted in a short lag



(a)



(b)

Figure 8. Rainfall-runoff responses during (a) the autumn season and (b) the spring season.

time between the rainfall and the runoff responses. The expansion of the middle zone during the spring period can be explained by higher vegetation control but less topographic control than that for the autumn. However, the development of fine roots during the spring season may decrease the water-storing capacity of the storage zone, and may result in faster release of water through interflow and higher peak runoff than those for the autumn season. The difference in the soil moisture recession patterns (Figures 3e

Table 4. Multiple Regression Analysis Statistics^a

CLS	6–22 November 2003								2–22 May 2004							
	10 cm Depth				30 cm Depth				10 cm Depth				30 cm Depth			
	TOT	UPP	MID	STO	TOT	UPP	MID	STO	TOT	UPP	MID	STO	TOT	UPP	MID	STO
ND	33	12	12	9	33	12	12	9	60	6	45	9	60	6	45	9
MRC	0.83	0.85	0.89	0.93	0.57	0.65	0.82	0.68	0.71	0.66	0.69	0.95	0.43	0.63	0.52	0.64
AR2	0.67	0.67	0.67	0.81	0.28	0.30	0.54	0.28	0.49	0.04	0.46	0.88	0.15	0.01	0.24	0.21
SE	2.83	3.31	1.04	2.46	4.81	4.15	0.89	6.84	1.70	2.06	1.36	0.97	2.18	1.36	1.53	2.91
R2PN	0.85	0.92	0.96	0.80	0.72	0.89	0.95	0.81	0.88	0.96	0.91	0.94	0.81	0.91	0.81	0.96

^aStatistics are with $y = ax_1 + bx_2 + c$, where y is the soil moisture recharge in percent; x_1 is the initial moisture content in percent; x_2 is the rainfall amount in mm; and a , b , and c are the regression coefficients. CLS indicates the classification of soil moisture responses; TOT = UPP + MID + STO; UPP, the upper zone; MID, the middle zone; STO, the storage zone. ND indicates the number of data. MRC indicates the multiple regression coefficient. AR2 indicates the adjusted coefficient of determination. SE indicates the standard error. R2PN indicates the coefficient of determination between predicted y and normal distribution.

and 3f) also indicates seasonally distinct roles of the storage zone to streamflow generation. A similar runoff response was found in a granitic unchanneled catchment in Japan [Haga *et al.*, 2005]. Considering the similar degrees of antecedent and mean saturation, seasonal differences in runoff responses can be explained by the influence of water conductance and storage distribution, as well as ecohydrological processes such as transpiration and interception. The greater influence of vegetation is associated with the longer soil moisture lag time and a smaller contributing area with quicker runoff response during the spring period than those for the autumn period. This fact was established regardless of similar saturation and rainfall input.

5. Conclusions

[28] The soil moisture response patterns for two selected periods were monitored and analyzed at a steep hillslope located in a mountainous forested watershed. On the basis of the temporal responses of the soil moisture at different locations for sequential rainfall events, soil moisture variation along the hillslope was characterized and categorized into three zones. The first, the upper zone, was located at the top of the hillslope where the soil moisture variation was primarily determined by rainfall. The second, the middle zone or the middle part of the hillslope, showed stable responses to rainfall events in both seasons. Finally, the storage zone represented the storage of soil water during the autumn while the recharge function was substantially diminished in the spring. During the spring, the size of the middle zone was increased.

[29] Given the similarity in rainfalls and mean saturation conditions between the two seasons, a faster soil moisture response, slower response of streamflow, and more variable distribution in point-scale soil moisture were observed during the autumn period than for the spring period. The LAI and meteorological measurements indicated higher interception and PET for the spring period. The correlation between the soil moisture mean and the contributing area was significant only near the bedrock during the autumn period. The recharge dominant characteristics of soil moisture for the autumn period changed into a less variable response pattern in the spring. Better predictability of soil moisture recharge was another characteristic of the autumn period. The substantial differences in seasonal runoff responses were explained by differences in soil moisture distribution and its cumulative water storage variation. Seasonal differences in potential evapotranspiration, canopy storage, water conductance, and soil water storage are responsible for the difference in soil moisture statistics, topographic correlation, water storage, and lag time for the soil moisture peak.

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